

Appendix 1 – Schedule of Payments

Contractor shall submit the monthly Statement under Sub-Clause 14.3 [Application of Interim Payment]. The monthly estimated contract value of the Works executed for the purpose of sub-paragraph (i) of Sub-Clause 14.3 [Application of Interim Payment] shall be valued based upon the payment milestone achieved in the corresponding month. “Approved” mentioned hereunder refer to “Approved by the Employer”.

The Sr. No. mentioned against each of the described payment milestone corresponds to the details of the same Sr. No. as mentioned under Price Schedule/ Tender Forms.

Description of payment milestone	Payment percentage / price
	(PKR, FC1, FC2, FC3)
Schedule of Rates and Prices: Schedule No. 2: Detailed Design Phase including Field Investigations and Capacity building	
2.1.1. Inception Report Approved 2.1.2 Liasion with stakeholders for NOC 2.1.3 Design Team mobilized	Sum of the prices in relevant currencies for Schedule No. 2 not exceeding 10% of the provision of Detailed Design Services in Schedule No. 2 of Tender Form in total for this milestone.
2.2.1. Design Review Report Approved 2.2.2. Required equipment made available at site for field investigations including field investigations conducted like Topographic Survey, Geo-technical Investigations. Drone Survey, GPR etc as mentioned in Sub Clause 4.1 of Employers' Requirements 2.2.3. Field Investigation Teams mobilized.	Sum of the prices in relevant currencies for Schedule No. 2 not exceeding 5% of the provision of Design Services in Schedule No. 2 of Tender Form in total for this milestone.
2.3.1. Field Investigation Report (Topographic, Geotech, survey, GPR etc.) Approved 2.3.2. Pre-Order inspection of MTBM Manufacturing Plant	Sum of the prices in relevant currencies for Schedule No. 2 not exceeding 10% of the provision of Design Services in Schedule No. 2 of Tender Form in total for this milestone.
2.4.1. Stage 1 Detailed Design Report Approved 2.4.2. Six (6) New MTBM Order(s) Placed 2.4.3 Pipes Manufacturing facility established	Sum of the prices in relevant currencies for Schedule No. 2 not exceeding 10% of the provision of Design Services in Schedule No. 2 of Tender Form in total for this milestone.
2.5.1. Stage 2 Detailed Design Report Approved 2.5.2. All Plans of Stage 2 Approved 2.5.3. Pumps Order Placed 2.5.4. Project Office Constructed for PMU 2.5.5 Pipes manufacturing started	Sum of the prices in relevant currencies for Schedule No. 2 not exceeding 20% of the provision of Design Services in Schedule No. 2 of Tender Form in total for this milestone.
2.6.1. E&S Documents Updated & approved including all Plans of E&S Approved 2.6.2. Work Plan & Construction Stage Plan Approved 2.6.3. Applicable E&S Complianaces made 2.6.4. Post-Order Inspection of MTBM Manufacturing Plant made 2.6.5. Capacity Building activities conducted	Sum of the prices in relevant currencies for Schedule No. 2 not exceeding 10% of the provision of Design Services in Schedule No. 2 of Tender Form in total for this milestone.
2.7.1. For each Site-specific Construction Drawings Approved on monthly basis 2.7.2. Applicable E&S Complianaces made 2.7.3. Monthly Progress Report Approved	Sum of proportionate prices in relevant currencies out of 30% of Design Services Cost for Schedule No. 2 of Tender Forms. The proportionate price will be calculated by

2.8. Field Investigations during Construction Phase	multiplying prices with %age of Physical Progress of construction work done approved in monthly progress reports. (30% of Design Services Cost for Schedule 2 of Tender form shall be paid on the basis of actual physical progress as per percentage completion of work done in the approved monthly progress report on account of services rendered of detailed design during construction period)
2.9. As Built Drawings Approved 2.10. O&M Manual Approved 2.11. Provisions of Licensed Softwares 2.12. Training and Capacity Building	Sum of the prices in relevant currencies for Schedule No. 2 not exceeding 5% of the provision of Design Services in Schedule No. 2 of Tender Form in total for this milestone.
2.13. Any Other item, if any, as specified in Schedule No. 2	Sum of proportionate prices in relevant currencies for item No. 2.13 of schedule 2 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of Physical Progress of respective construction work approved in monthly progress reports.
Schedule No. 3: Civil Works	
3.1. Conditions Survey of buildings (Crack Survey) within zone of influence along the route of laying of pipes through microtunnelling before laying and after completion of laying of pipes, implementation of remedial measures, (if any), etc., completed in all respects and Approved 3.2. Vibration Analysis & monitoring of sensitive receptors i.e. school, over head & underground nearby structures, sensitive buildings, vulnerable structures etc.	Sum of proportionate prices in relevant currencies of item No. 3.1 & 3.2 under schedule 3 of tender forms. The proportionate prices will be calculated by multiplying prices with number of buildings surveyed out of of total buildings falling under zone of influence
3.3. For each completed segment of 500 m of RCC Pipe of dia 1050mm laid with MTBM in a continous reach in each month, along with Construction of Jacking & Receiving shafts, compacted back filling of jacking & receiving shafts, Completed in all respect.	Sum of proportionate prices in relevant currencies of item No. 3.3 under schedule 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of total length of segments of pipeline laid w.r.t total length of 1050mm pipeline in scope.
3.4. For each completed segment of 250 m of RCC Pipe of dia 1500mm laid with MTBM in a continous reach in each month, along with Construction of Jacking & Receiving shafts, compacted back filling of jacking & receiving shafts, Completed in all respect.	Sum of proportionate prices in relevant currencies of item No.3.4 under schedule 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of length of segments of pipeline laid w.r.t total length of 1500mm pipeline in scope.
3.5. For each completed segment of 500 m of RCC Pipe of dia 2400mm laid with MTBM in a continous reach in each month, along with Construction of Jacking & Receiving shafts, compacted back filling of jacking & receiving shafts, Completed in all respect.	Sum of proportionate prices in relevant currencies of item No.3.5 under schedule 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of length of segments of pipeline laid w.r.t total length of 2400mm pipeline in scope.
3.6. For each completed segment of 250 m of RCC Pipe of dia 3500mm laid with MTBM in a continous reach in each month, along with Construction of Jacking & Receiving shafts, compacted back filling of jacking & receiving shafts, Completed in all respect including widening of Access Road along	Sum of proportionate prices in relevant currencies of items No.3.6 under schedule 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of length of segments of pipeline laid w.r.t total length of 3500mm pipeline in scope.

Cantt drain its restoration after work, Access road path for mobility to Camp sites and Dewatering	
3.7.1 For each Constructed RCC Manholes on 1050mm sewer line in each month complete in all respects, including compacted back filling, steps, benching, RCC top slab, manhole covers etc.	Sum of proportionate prices in relevant currencies for item No. 3.7.1 under schedule 3 of the tender forms. The proportionate prices will be calculated by multiplying prices with %age of total number of Manholes constructed on monthly basis with respect to total number of Manholes on 1050mm in scope sewer line
3.7.2. For each Constructed RCC Manholes on 1500mm sewer line in each month complete in all respects, including compacted back filling, steps, benching, RCC top slab, manhole covers etc.	Sum of proportionate prices in relevant currencies for item No. 3.7.2 under schedule 3 of the tender forms. The proportionate prices will be calculated by multiplying prices with %age of total number of Manholes constructed on monthly basis with respect to total number of Manholes on 1500mm line
3.7.3. For each Constructed RCC Manholes on 2400mm sewer line in each month complete in all respects, including compacted back filling, steps, benching, RCC top slab, manhole covers etc.	Sum of proportionate prices in relevant currencies for item No. 3.7.3 under schedule 3 of the tender forms. The proportionate prices will be calculated by multiplying prices with %age of total number of Manholes constructed on monthly basis with respect to total number of Manholes on 2400mm line
3.7.4. For each Constructed RCC Manholes on 3500mm sewer line in each month complete in all respects, including compacted back filling, steps, benching, RCC top slab, manhole covers etc.	Sum of proportionate prices in relevant currencies for items No. 3.7.4 under schedule 3 of the tender forms. The proportionate prices will be calculated by multiplying prices with %age of total number of Manholes constructed on monthly basis with respect to total number of Manholes on 3500mm line.
3.8 For each completion of Reconstruction (Patchwork) of dismantled roads/tiles/PCC site along manhole/ jacking & receiving shafts/ other sites in each month etc at all excavated locations dismantled for construction. Re-instate to original conditions for public/ private properties	Sum of proportionate prices in relevant currencies for items No. 3.8 under schedule 3 of the tender forms. The proportionate prices will be calculated by multiplying prices with %age of re-constructed sites w.r.t total reconstruction work of shafts/manholes.
3.9 For each constructed Intake structure in each month, complete in all respect and connected with existing system (sizes and dimension in Employer Requirements) at premises of existing intermittent pumping stations and construction of by-pass arrangement with penstock gates to ensure diversion of emergency flow into existing pumping station	Sum of proportionate prices in relevant currencies for items No. 3.9 under schedule 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of cusec capacity of pumping stations constructed/ of which flow catered into system on monthly basis with respect to total discharge capacity (i.e. 688 cfs)
3.10 For each constructed sewerage system 5000 PSI), alongwith required manholes, by steel casing / manual excavation, including bypass arrangement, making connection of existing sewerage system with new intake structures involving dewatering, provision of temporary arrangements for disposal of sewage waste	Sum of proportionate prices in relevant currencies for item No. 3.10 of schedule 3 of tender forms. The proportionate prices will be calculated by multiplying prices with % of length of pipes laid completely at any pumping station with respect to total length of bypass sewer to be laid for all pumping stations (23 No)
3.11. Control rooms (sizes and dimension in Employer Requirements) at drop structures in premises of existing disposal stations including excavation of soil for foundation,	Sum of proportionate prices in relevant currencies for item no 3.11 of Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with number of

construction of foundation slab with reinforced concrete, installation of embedded parts for control units, construction of walls, reinforced concrete roof, installation of doors including locks	control room constructed with respect to total number of control rooms to be constructed for drop structures (16 No)
3.12. For each constructed drop structures at inlet structures in each month, complete in all respects	Sum of proportionate prices in relevant currencies for Item No. 3.12 in Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of number of constructed Drop structures on monthly basis with respect to total number of Drop Structures to be constructed (16 No)
3.13. For each constructed drop manhole at inlet structures in each month, complete in all respects	Sum of proportionate prices in relevant currencies for Item No. 3.13 in Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of number of drop manholes constructed on monthly basis with respect to total number of Drop manholes to be constructed at intakes in scope (i.e.7 No)
3.14. For each generator replaced and made functional in each month, complete in all respects	Sum of proportionate prices in relevant currencies for Item No. 3.14 in Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of number of generators replaced on monthly basis with respect to total number of generators replaced
3.15 For each Control Panel Rooms constructed/ replaced in each month, complete in all respects	Sum of proportionate prices in relevant currencies for Item No. 3.15 in Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of total number of control panel room constructed on monthly basis with respect to total number of control panel rooms (11 No)
3.16. For each Building Room dismantled in each month, complete in all respects	Sum of proportionate prices in relevant currencies for Item No. 3.16 in Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of total number of building room dismantled on monthly basis with respect to total number of building room in scope
3.17. For each Lift station/ intermittent pumping station revamped/ renovated in each month, complete in all respects	Sum of proportionate prices in relevant currencies for Item No. 3.17 in Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of number of lift stations revamped on monthly basis with respect to total number of lift stations revamped
3.18. Gratings installed at each site on each month, complete in all respect	Sum of proportionate prices in relevant currencies for Item No. 3.18 in Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of total number of sites completed on monthly basis with respect to total number of sites
3.19. For construction of each control room on each month along lines and laterals, complete in all respects	Sum of proportionate prices in relevant currencies for Item No. 3.19 in Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of number of control rooms completed on monthly

	basis with respect to total number of control rooms constructed
3.20 Drop structures constructed at Qurtaba chowk along Line A and Line B near Qurtaba chowk	Sum of proportionate prices in relevant currencies for Item No. 3.20 in Schedule No. 3 of tender forms. The proportionate prices will be calculated against each of completed drop structure after completion
3.21 Storm water inundation control room along with gratings and drop manholes,	Sum of proportionate prices in relevant currencies for Item No. 3.21 in Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of number of inundation control completed on monthly basis with respect to total number of storm water inundation control to be constructed
3.22 For dismantling & site clearance of each building at Gulshan-e-Ravi Disposal Station on each month, including multiple office buildings, generator house, transformer building, court complete in all respect.	Sum of proportionate prices in relevant currencies for Item No. 3.22 in Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of number of Buildings dismantled on monthly basis with respect to total number of buildings dismantled
3.23. Generator House constructed, complete in all respect	Sum of the prices in relevant currencies for item No. 3.23 of schedule 3 of Tender Form in total for this milestone
3.24. Transformer Building constructed, complete in all respect	Sum of the prices in relevant currencies for item No. 3.24 of schedule 3 of Tender Form in total for this milestone
3.25. RCC Channel dismantled, including disposal arrangement of pumped sewage to spill channel and diversion of flows toward new channel after construction, complete in all respects.	10% of the Sum of the prices in relevant currencies for item No. 3.25 of schedule 3 of Tender Form for excavation. 20% of the Sum of the prices in relevant currencies for item No. 3.25 of schedule 3 of Tender Form on dismantling of top slab. 40% of the Sum of the prices in relevant currencies for item No. 3.25 of Tender Form on dismantling of side walls. 20% of the Sum of the prices in relevant currencies for Schedule No. 3.25 of Tender Form on dismantling of bed 10% of the Sum of the prices in relevant currencies for Schedule No. 3.25 of Tender Form on site clearance
3.26. Dismantled Head Regulator of Outfall Drain complete in all respects.	Sum of the prices in relevant currencies for Schedule No. 3.26 of head regulator of outfall drain
3.27. Construction of fully furnished three-story Office Building in frame structure, each story having covered area of at least 500 Square meter	Sum of Proportions of prices as mentioned in relevant currencies for Schedule No. 3.27 of Tender Form upon its completion
3.28 SCADA Control Room Constructed	Sum of Proportion of prices in relevant currencies for Schedule No. 3.28 of Tender Form upon its completion

3.29.1 Construction of Administrative Buildings for Disposal staff new & existing	Sum of Proportion of the prices in relevant currencies for Schedule No. 3.29.1 of Tender Form upon its completion
3.29.2. Solar Room	Proportion of the prices in relevant currencies for Schedule No. 3.29.2 of Tender Form upon its completion
3.29.3 State of the Art Project Demonstration Hall	Proportion of the prices in relevant currencies for Schedule No. 3.29.3 of Tender Form upon its completion
3.29.4. Construction of Administrative Buildings for Displaced staff	Sum of the prices in relevant currencies for Schedule No. 3.29.4 of Tender Form upon its completion
3.29.5 Meeting room	Sum of the prices in relevant currencies for Schedule No. 3.29.5 of Tender Form in total for this milestone
3.30. Constructed Outfall Concrete Channel of Existing Pump House complete in all respects	Sum of the prices in relevant currencies for Schedule No. 3.30 of Tender Form upon its completion
3.31.1 Intake Structure of Ultimate Disposal Station Constructed	10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.1 of Tender Form for Sheet piling & excavation. 10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.1 of Tender Form on completion of foundation & bed. 40% of the Sum of the prices in relevant currencies for Schedule No. 3.31.1 of Tender Form for steel reinforcement. 30% of the Sum of the prices in relevant currencies for Schedule No. 3.31.1 of Tender Form for concrete filling till top of the intake structure. 10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.1 of Tender Form on completion of intake structures.
3.31.2 Screening Chamber of Ultimate Disposal Station Constructed	10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.2 of Tender Form for Sheet piling & excavation. 10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.2 of Tender Form on completion of foundation & bed. 40% of the Sum of the prices in relevant currencies for Schedule No. 3.31.2 of Tender Form for steel reinforcement. 30% of the Sum of the prices in relevant currencies for Schedule No. 3.31.2 of Tender Form for concrete filling till top of the structure 10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.2 of Tender Form on completion of structure.

3.31.3 Wet well of Ultimate Disposal Station Constructed	10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.3 of Tender Form for Sheet piling & excavation. 10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.3 of Tender Form on completion of foundation & bed. 40% of the Sum of the prices in relevant currencies for Schedule No. 3.31.3 of Tender Form for steel reinforcement. 30% of the Sum of the prices in relevant currencies for Schedule No. 3.31.3 of Tender Form for concrete filling till top of the structure 10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.3 of Tender Form on completion of structure.
3.31.4 Dry Well of Ultimate Disposal Station Constructed	10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.4 of Tender Form for Sheet piling & excavation. 10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.4 of Tender Form on completion of foundation & bed. 40% of the Sum of the prices in relevant currencies for Schedule No. 3.31.4 of Tender Form for steel reinforcement. 30% of the Sum of the prices in relevant currencies for Schedule No. 3.31.4 of Tender Form for concrete filling till top of the structure 10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.4 of Tender Form on completion of structure
3.31.5 Pump House Building of Ultimate Disposal Station Constructed	10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.5 of Tender Form for foundation. 40% of the Sum of the prices in relevant currencies for Schedule No. 3.31.5 of Tender Form for retaining walls. 40% of the Sum of the prices in relevant currencies for Schedule No. 3.31.5 of Tender Form roof. 10% of the Sum of the prices in relevant currencies for Schedule No. 3.31.5 of Tender Form on completion of structure.
3.31.6 Concrete Channel up to distribution chamber of Ultimate Disposal Station Constructed	Sum of the prices in relevant currencies for Schedule No. 3.31.6 of Tender Form in total for this milestone

3.31.7 Site restored & beautification completed with tree plantation & horticulture at all lift stations & ultimate disposal station etc	Sum of the prices in relevant currencies for Schedule No. 3.31.7 of Tender Form in total for this milestone
3.31.8 Supply, installation, testing, and commissioning including but not limited to all types of Steel Grating with Frames and Safety Railings, platforms, walkways, ladders, manholes, covers, Chemical/Mechanical Anchors	Sum of the prices in relevant currencies for Schedule No. 3.31.8 of Tender Form in total for this milestone
3.32. Construction of Distribution Chamber complete in all respects.	10% of the Sum of the prices in relevant currencies for Schedule No. 3.32 of Tender Form for Sheet piling & excavation. 10% of the Sum of the prices in relevant currencies for Schedule No. 3.32 of Tender Form on completion of foundation & bed. 40% of the Sum of the prices in relevant currencies for Schedule No. 3.32 of Tender Form for steel reinforcement. 30% of the Sum of the prices in relevant currencies for Schedule No. 3.32 of Tender Form for concrete filling till top of the structure 10% of the Sum of the prices in relevant currencies for Schedule No. 3.32 of Tender Form on completion of structure
3.33. Dismantling of Existing Reinforced Concrete Channel from Distribution Chamber to Outfall Disposal Drain including foundations and dispose off material at designated place. Disposal arrangement of pumped sewage to spill channel and diversion of flows toward new channel after construction	10% of the Sum of the prices in relevant currencies for Schedule No. 3.33 of Tender Form for excavation. 20% of the Sum of the prices in relevant currencies for Schedule No. 3.33 of Tender Form on dismantling of top slab. 40% of the Sum of the prices in relevant currencies for Schedule No. 3.33 of Tender Form on dismantling of side walls. 20% of the Sum of the prices in relevant currencies for Schedule No. 3.33 of Tender Form on dismantling of bed 10% of the Sum of the prices in relevant currencies for Schedule No. 3.33 of Tender Form on site clearance
3.34. Construction of Outfall Concrete Conduit, reconstruction of road dismantled during construction, restoration of surrounding roads, etc. complete in all respects.	10% of the Sum of the prices in relevant currencies for Schedule No. 3.34 of Tender Form for excavation. 20% of the Sum of the prices in relevant currencies for Schedule No. 3.34 of Tender Form on dismantling of top slab.

	40% of the Sum of the prices in relevant currencies for Schedule No. 3.34 of Tender Form on dismantling of side walls. 20% of the Sum of the prices in relevant currencies for Schedule No. 3.34 of Tender Form on dismantling of bed 10% of the Sum of the prices in relevant currencies for Schedule No. 4.34 of Tender Form on site clearance
3.35. Rehabilitation by excavation/ dredging from bed and banks in flowing sewage and Construction of Brick walls along both banks of Disposal Outfall Drain, disposal of excavated material, backfill, restoration of roads and plantation along both banks between roads and brick walls, complete in all respects.	10% of the Sum of the prices in relevant currencies for Schedule No. 3.35 of Tender Form for excavation & cross-section widening. 10% of the Sum of the prices in relevant currencies for Schedule No. 3.35 of Tender Form on water diversion. 20% of the Sum of the prices in relevant currencies for Schedule No. 3.35 of Tender Form on construction of access. 20% of the Sum of the prices in relevant currencies for Schedule No. 3.35 of Tender Form on construction of foundation wall. 30% of the Sum of the prices in relevant currencies for Schedule No. 3.35 of Tender Form on construction of retaining wall. 10% of the Sum of the prices in relevant currencies for Schedule No. 3.35 of Tender Form on site clearance.
3.36. Construction of Head Regulator of Disposal Outfall Drain and restoration of road along both side of the head regulator complete in all respects.	10% of the Sum of the prices in relevant currencies for Schedule No. 3.36 of Tender Form for excavation & cross-section widening. 20% of the Sum of the prices in relevant currencies for Schedule No. 3.36 of Tender Form on water diversion & dismantling existing structure. 30% of the Sum of the prices in relevant currencies for Schedule No. 3.36 of Tender Form on construction of access road. 30% of the Sum of the prices in relevant currencies for Schedule No. 3.36 of Tender Form on construction of head regulator. 10% of the Sum of the prices in relevant currencies for Schedule No. 3.36 of Tender Form on operation of head regulator.

3.37. Construction of Head Regulator of Spill Channel/escape diversion channel, complete in all respects.	10% of the Sum of the prices in relevant currencies for Schedule No. 3.37 of Tender Form for excavation & cross-section widening. 20% of the Sum of the prices in relevant currencies for Schedule No. 3.37 of Tender Form on water diversion & dismantling existing structure. 50% of the Sum of the prices in relevant currencies for Schedule No. 3.37 of Tender Form on construction of head regulator. 20% of the Sum of the prices in relevant currencies for Schedule No. 3.37 of Tender Form on operation of head regulator.
3.38. For each segment of 500m of drains rehabilitated on monthly basis complete in all respect	Sum of proportionate prices in relevant currencies for Item No. 3.38 in Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of segments of drains rehabilitated on monthly basis with respect to total drain length
3.39. For each segment of 500m of rider sewer laid with manholes on monthly basis complete in all respect	Sum of proportionate prices in relevant currencies for Item No. 3.39 in Schedule No. 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of number of rider sewer laid on monthly basis with respect to total sewer length in scope
3.40. For each excavated site, services shifted/ by-passed on temporary/ permanent basis and restoration of services on each month complete in all respect.	Sum of proportionate prices in relevant currencies of item No.3.40.1 in schedule 3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of gas lines shifted w.r.t total quantity of gas line proposed to be shifted as per scope. Sum of proportionate prices in relevant currencies of Schedule No.3.40.2 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of length of electric lines shifted w.r.t total length proposed to be shifted as per scope. Sum of proportionate prices in relevant currencies of Schedule No.3.40.3 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of length of water supply and sewerage lines are shifted w.r.t total length of water supply and sewerage lines effected/proposed to be shifted as per scope Sum of proportionate prices in relevant currencies of Schedule No.3.40.4 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of length of telephone & internet services including fiber cables are shifted w.r.t total length of telephone and internet services i/c fibre cables

	Sum of proportionate prices in relevant currencies of Schedule No.3.40.5 of tender forms. The proportionate prices will be calculated by multiplying prices with %age of length of sites at which PSCA & PHA services are shifted w.r.t total no. of sites in scope.
3.41. Any other item	Sum of proportionate prices in relevant currencies for item No. 3.41 of schedule 3 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of Physical Progress of respective construction work approved in monthly progress reports.
Note: The progress payment against civil work shall be measured, estimated and approved by the Engineer against the total quantities for excavation, concrete, steel, backfill, building works, etc. related to each structure (provided by the Contractor after detailed design). Such payment shall be duly approved on the basis of monthly approved progress report and paid according to the milestones mentioned for each structure/ civil work.	
Schedule No. 4: Mechanical Works	
4.1. Completion of Mechanical Works on inlet structures as per 4.1 of the tender forms, at each Lift Station on each month, complete in all respect	Sum of proportionate prices in relevant currencies for item No. 4.1 of schedule 4 of Tender Forms. The proportionate price of mechanical work on inlet structures will be calculated by multiplying prices with %age of capacity in cusec of completed sites w.r.t total capacity in cusec for all sites in scope.
4.2. Completion of screening system at each Lift Station on each month, complete in all respect	Sum of proportionate prices in relevant currencies for item No. 4.2 of schedule 4 Tender Forms. The proportionate price will be calculated by multiplying prices with %age of capacity in cusec of completed sites w.r.t total capacity in cusec for all sites in scope. 70% of proportion of each site on delivery of complete equipment 30% of proportion of each site on installation of complete equipment
4.3. Completion of ventilation & odor control system at each Lift Station on each month, complete in all respect	Sum of proportionate prices in relevant currencies for item No. 4.3 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of site of ventilation & odor control system of completed sites w.r.t total number of sites in scope. 70% of proportion of each site on delivery of complete equipment 30% of proportion of each site on installation of complete equipment
4.4 Completion of ventilation & odor control system along sewerlines	Sum of proportionate prices in relevant currencies for item No. 4.4 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of site of ventilation & odor control system completed w.r.t total number of sites in scope. 70% of proportion of each site on delivery of complete equipment

	30% of proportion of each site on installation of complete equipment
4.5. Screening disposal trucks to accommodate screening materials of all eliminating pumping stations	Sum of proportionate prices in relevant currencies for item No. 4.5 of schedule 4 of Tender Forms. The proportionate price will be calculated by number of trucks delivered w.r.t total number of proposed trucks to be delivered (10 No.) as per specifications.
4.6. Completion of mechanical works at inlet structure of ultimate disposal station, complete is all respect	70 % of the Sum of prices in relevant currencies for item No. 4.6 of schedule 4 of Tender Forms at the time of delivery of all items 30 % of the Sum of prices in relevant currencies for item No. 4.6 of schedule 4 Tender Forms at the time of installation of all items
4.7. Completion of screening system of ultimate disposal station, complete is all respect	70 % of the Sum of prices in relevant currencies for item No 4.7 of schedule 4 of Tender Forms at the time of delivery of all items 30 % of the Sum of prices in relevant currencies for Schedule No. 4.7 of Tender Forms at the time of installation of all items
4.8. Installation, testing & commission of Pumps & Motors and allied works at ultimate disposal station	70 % of the Sum of prices in relevant currencies for item No. 4.8 of schedule 4 of Tender Forms at the time of delivery of all items 30 % of the Sum of prices in relevant currencies for item No. 4.8 of schedule 4 of Tender Forms at the time of installation of all items Sum of prices in relevant
4.9. Hydraulic modeling of pumping station & drop structures performed	Sum of prices in relevant currencies for item No. 4.9 of schedule 4 of Tender Forms.
4.10 Installed all Pipe & Fittings (Supply, installation, testing, and commissioning of all types of Pipes & Fittings) complete in all respects.	Sum of prices in relevant currencies for item No. 4.10 of schedule 4 of Tender Forms.
4.11. Installed all Valves (Supply, installation, testing, and commissioning of all kinds of Valves) complete in all respects.	Sum of prices in relevant currencies for item No. 4.11 of schedule 4 of Tender Forms.
4.12. Accessories (Supply, installation, testing, and commissioning of Automatic Overhead Travelling Cranes, Submersible Sump Pumps, lifting arrangement, all types of Steel Grating with Frames and Safety Railings, platforms, walkways, ladders, manholes, covers, Chemical/Mechanical Anchors, Protection coating, paints etc. with all accessories like Nuts, Bolts, Washers, Gaskets, safety of equipment and persons, storage, all kind of labour/manpower, consumables, utilities, power, etc.) under Item No. 5.158, complete in all respects.	Sum of prices in relevant currencies for item No. 4.12 of schedule 4 of Tender Forms.
4.13. Delivered sewer inspection equipment for O&M of Sewer system & Trained staff to use the equipment	Sum of prices in relevant currencies for item No. 4.13 of schedule 4 of Tender Forms.
4.14. Delivered Sewer cleaning robot for O&M of sewer system & trained staff to use the equipment	Sum of prices in relevant currencies for item No. 4.14 of schedule 4 of Tender Forms.
4.15. Delivered Sewer cleaning heavy duty jetting units (3 no.) for O&M of sewer system & trained staff to use the equipment	Sum of prices in relevant currencies for item No. 4.15 of schedule 4 of Tender Forms.

4.16. Delivered Sewer cleaning heavy duty suction units (3 no.) for O&M of sewer system & trained staff to use the equipment	Sum of prices in relevant currencies for item No. 4.16 of schedule 4 of Tender Forms.
4.17. Any other item	Sum of proportionate prices in relevant currencies for item No. 4.17 of schedule 4 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of Physical Progress of respective construction work approved in monthly progress reports.
Schedule No. 5: Electrical Works	
5.1. Installed Low Voltage Switchgears at each intake structure for screen system, Ventilation & Order Control and allied system each month, complete in all respect.	Sum of proportionate prices in relevant currencies for item No. 5.1 of schedule 5 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of capacity in cusec of completed sites w.r.t total capacity in cusec for all sites in scope. 70% of proportion of each site on delivery of complete equipment 30% of proportion of each site on installation of complete equipment
5.2-5.7. Electrification of each intake structure, control panel, generators, control panels each month, including but not limited to installation of Low voltage cables, cable containment arrangement, earthing, lightening installation, MV Cable installation, small power equipment & installation etc. complete in all respect	Sum of proportionate prices in relevant currencies for items No. 5.2,5.3,5.4,5.5,5.6 & 5.7 of schedule 5 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of number of intake sites completed w.r.t total number of intake sites in scope.
5.8-5.13. Electrification of each control room & control points, control panel, generators, including but not limited to installation of Low voltage switchgear & cables, cable containment arrangement, earthing, lightening installation, small power equipment & installation etc. complete in all respect	Sum of proportionate prices in relevant currencies for items No. 5.8,5.9,5.10,5.11,5.12 & 5.13 of schedule 5 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of number of control room sites completed w.r.t total number of control room sites in scope.
5.14-5.18 Electrification of each control room & control points, control panel, generators, Low voltage switch gear including but not limited to installation of Low voltage cables, cable containment arrangement, earthing, lightening installation, small power equipment & installation etc. complete in all respect	Sum of proportionate prices in relevant currencies for items No. 5.14,5.15,5.16,5.17 & 5.18 of schedule 5 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of number of control point sites completed w.r.t total number of control point sites in scope.
5.19. Installed Medium Voltage Switchgears at ultimate disposal station and allied system, complete in all respect.	70 % of the Sum of prices in relevant currencies for items No. 5.19 of schedule 5 of Tender Forms at the time of delivery of all items 30 % of the Sum of prices in relevant currencies for items No. 5.19 of schedule 5 of Tender Forms at the time of installation of all items
5.20. Installed MV Synchronization & Load Sharing Panels at ultimate disposal station and allied system, complete in all respect.	70 % of the Sum of prices in relevant currencies for item No 5.20 of schedule 5 of Tender Forms at the time of delivery of all items 30 % of the Sum of prices in relevant currencies for item No. 5.20 of schedule 5 of Tender Forms at the time of installation of all items

5.21. Installed MCUs, VFDs, MV Synchronization & Load Sharing Panels at ultimate disposal station and allied system, complete in all respect.	70 % of the Sum of prices in relevant currencies for item No. 5.21 of schedule 5 of Tender Forms at the time of delivery of all items 30 % of the Sum of prices in relevant currencies for item No. 5.21 of schedule 5 of Tender Forms at the time of installation of all items
5.22. Installed MDBs, SDBs, MCCs, MV Synchronization & Load Sharing Panels at ultimate disposal station and allied system, complete in all respect.	70 % of the Sum of prices in relevant currencies for item No. 5.22 of schedule 5 of Tender Forms at the time of delivery of all items 30 % of the Sum of prices in relevant currencies for item No. 5.22 of schedule 5 of Tender Forms at the time of installation of all items
5.23. Installed LV Synchronization & Load Sharing Panels at ultimate disposal station and allied system, complete in all respect.	70 % of the Sum of prices in relevant currencies for item No. 5.23 of schedule 5 of Tender Forms at the time of delivery of all items 30 % of the Sum of prices in relevant currencies for item No. 5.23 of schedule 5 of Tender Forms at the time of installation of all items
5.24. Installed MV Transformer at ultimate disposal station and allied system, complete in all respect.	70 % of the Sum of prices in relevant currencies for item No. 5.24 of schedule 5 of Tender Forms at the time of delivery of all items 30 % of the Sum of prices in relevant currencies for item No. 5.24 of schedule 5 of Tender Forms at the time of installation of all items
5.25. Installed Low Voltage Transformer at ultimate disposal station and allied system, complete in all respect.	70 % of the Sum of prices in relevant currencies for item No. 5.25 of schedule 5 of Tender Forms at the time of delivery of all items 30 % of the Sum of prices in relevant currencies for item No. 5.25 of schedule 5 of Tender Forms at the time of installation of all items
5.26. Installed Medium Voltage Power Generators at ultimate disposal station and allied system, complete in all respect.	70 % of the Sum of prices in relevant currencies for item No. 5.26 of schedule 5 of Tender Forms at the time of delivery of all items 30 % of the Sum of prices in relevant currencies for items No. 5.26 of schedule 5 of Tender Forms at the time of installation of all items
5.27. Installed Low Voltage Power Generators at ultimate disposal station and allied system, complete in all respect.	70 % of the Sum of prices in relevant currencies for item No. 5.27 of schedule 5 of Tender Forms at the time of delivery of all items 30 % of the Sum of prices in relevant currencies for item No. 5.27 of schedule 5 of Tender Forms at the time of installation of all items
5.28-5.46. Installed 110VDC System at ultimate disposal station and allied system, complete in all respect. Electrical works completed at ultimate Disposal Station including but not limited to installation of Medium Voltage cables, Low Voltage cables, cable containers, UPS, earthing, Lightning Protection, Lightning Installation, Emergency Lighting, small power Equipment, EPABX, Computer	Sum of prices in relevant currencies for items No. 5.28,5.29,5.30,5.31,5.32,5.33,5.34,5.35, 5.36, 5.37, 5.38, 5.39, 5.40, 5.41, 5.42, 5.43, 5.44, 5.45, 5.46 of schedule 5 of Tender Forms. 70 % of the Sum of prices in relevant currencies for above items. of Tender Forms at the time of delivery of all items

Networking, Fire Alarm, Fire Suppression, CCTV installation & monitoring, Public Address System, Access control System, Walk Through gate, connections with panels, motors, transformers, generators, and all allied equipment to make Disposal Station Functional complete in all respect.	30 % of the Sum of prices in relevant currencies for above items of Tender Forms at the time of installation of all items
5.47. Installed 450kw Solar System at ultimate disposal station, complete in all respect.	75 % of the Sum of prices in relevant currencies for Schedule No. 5.47 of Tender Forms at the time of delivery of all items 25 % of the Sum of prices in relevant currencies for Schedule No. 5.47 of Tender Forms at the time of installation of all items
5.48. Electrical Works completed at buildings constructed as per ER, complete in all respect.	Sum of prices in relevant currencies for Schedule No. 5.48 of Tender Forms
5.49. Any other item	Sum of proportionate prices in relevant currencies for item No. 5.49 of schedule 5 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of Physical Progress of respective construction work approved in monthly progress reports.
Schedule No. 6: SCADA System Works	
6.1-6.3. Completely installed SCADA System Works at Disposal / Pumping Station, complete in all respects	75 % of the Sum of prices in relevant currencies for Schedule No. 6.1, 6.2 & 6.3 of Tender Forms at the time of delivery of all items 25 % of the Sum of prices in relevant currencies for Schedule No. 6.1,6.2 & 6.3 of Tender Forms at the time of installation of all items
6.4 Completely installed SCADA System Works at WASA Lahore Head Office, complete in all respects	Sum of prices in relevant currencies for Schedule No. 6.4 of Tender Forms
6.5 Completely installed SCADA control points on manholes along lines & laterals, complete in all respects	Sum of proportionate prices in relevant currencies for Schedule No. 6.5 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of control points established w.r.t total control points in scope of work..
6.6 Completely installed SCADA control points at each intake structure, complete in all respects	Sum of proportionate prices in relevant currencies for Schedule No. 6.6 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of intake sites completed w.r.t total intake sites in scope of work..
6.7. Any other item	Sum of proportionate prices in relevant currencies for item No. 6.8 of schedule 6 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of Physical Progress of respective construction work approved in monthly progress reports.
Schedule No. 7: Testing and Commissioning	
7.1-7.5. Successful Testing and Commissioning of entire facility at completion of Works, complete in all respects	Sum of prices in relevant currencies for items No. 7.1, 7.2,7.3,7.4,7.5,of schedule 7 of Tender Forms

7.6. Successful Testing and Commissioning of entire facility after completion of Works, complete in all respects	Sum of prices in relevant currencies for items No. 7.6 of schedule 7 of Tender Forms
7.7. Any other item	Sum of proportionate prices in relevant currencies for item No. 7.7 of schedule 7 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of Physical Progress of respective construction work approved in monthly progress reports.
Schedule No.8:Environmental, Social, Health & Safety (ESHS)	
8.1. Environmental Monitoring Completed each quarter of a year as per ER complete in all respects	Sum of proportionate prices in relevant currencies for Schedule No. 8.1 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of Physical Progress of respective construction work approved in monthly progress reports.
8.2. Social Monitoring Completed each month of a year as per ER complete in all respects	Sum of proportionate prices in relevant currencies for Schedule No. 8.2 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of Physical Progress of respective construction work approved in monthly progress reports.
8.3. OHS/CHS Completed each month of a year as per ER complete in all respects	Sum of proportionate prices in relevant currencies for Schedule No. 8.3 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of Physical Progress of respective construction work approved in monthly progress reports.
8.4-8.7. Cultural heritage protected, grievances redressed, & stake holders engagement conducted, Training & capacity Building sessions	Sum of proportionate prices in relevant currencies for Schedule No. 8.4, 8.5 & 8.6, 8.7 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of Physical Progress of respective construction work approved in monthly progress reports.
8.8. Completed ESHS Audits & filled O&M E&S Requirements 8.9. Operation and Maintenance	Sum of proportionate prices in relevant currencies for Schedule No. 8.8, 8.9 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of Physical Progress of respective construction work approved in monthly progress reports.
8.10. Any other item	Sum of proportionate prices in relevant currencies for item No. 8.10 of schedule 8 of Tender Forms. The proportionate price will be calculated by multiplying prices with %age of Physical Progress of respective construction work approved in monthly progress reports.

Percentage Discharge of Lift Stations			
Sr. No.	Disposal Station / Area	Sewage Flows 2050 (cfs)	Percentage (%)
1	Harbanspura	82.56	12.00
2	Mian Mir	9.01	1.31
3	Infantry Road DS	56.75	8.25
4	Bird Market DS	26.78	3.89
5	Larech Colony DS	2.08	0.30
6	Ghazi Mohallah DS	2.35	0.34
7	Muhammad Nagar DS	1.39	0.20
8	Governor House DS	9.65	1.40
9	Lakshami Chowk DS	82.73	12.02
10	Lytton Road DS	31.34	4.56
11	PIA Planetarium DS	20.97	3.05
12	Gurumangat Road DS	21.19	3.08
13	Q Block Gulberg DS	20.17	2.93
14	G Block Gulberg DS	35.95	5.23
15	Zafar Ali Road DS	4.09	0.59
16	Shareef Colony DS	1.69	0.25
17	Shadman DS	43.69	6.35
18	Shama DS	14.89	2.16
19	Ichra DS	25.63	3.73
20	Rasool Park DS	94.75	13.77
21	Poonch Road DS	75.07	10.91
22	Jhan ki Devi	0.46	0.07
23	Centre Point	24.81	3.61
Total Flows (cfs)		688	100.00